Théo Duez

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Born on 11 September 2000 – Based in Paris, France

Education

Oct 2024–Sep 2027
PhD in Applied Mathematics for Quantum Chemistry
CNRS – CERMICS, École Nationale des Ponts et Chaussées, Champs-sur-Marne, France
Thesis: New formulations, approximations and numerical methods for Time-Dependent Density Functional Theory (TDDFT)
Supervisors: Éric Cancès and Mathieu Lewin

Sep 2020 – Sep 2024
Master of Science (Diplôme d'Ingénieur), Mathematics specialization
CentraleSupélec - Gif-sur-Yvette
Includes a gap year (Sep 2022 – Sep 2023) for research internships at CEA Paris-Saclay and the Max Planck Institute – see Research Experience

Sep 2023 – Sep 2024
Master 2 (M2), Analysis, Modelling and Simulation (AMS)
Université Paris-Saclay
Completed concurrently with the final year at CentraleSupélec

Sep 2021 – Jul 2022
Master 1 (M1), Fundamental Mathematics track
Université Paris-Saclay
Completed concurrently with CentraleSupélec; two courses taken at Saclay, remaining credits validated from CentraleSupélec coursework

Sep 2020 – Jul 2021
Bachelor's Degree, Mathematics
Université Paris-Saclay
Completed concurrently with the first year at CentraleSupélec

Sep 2018 – Aug 2020
Classes Préparatoires (CPGE), MP – option Informatique
Collège Stanislas, Paris

Research Experience

Apr 2024 – Aug 2024
Research Intern
Inria Paris & CERMICS, Champs-sur-Marne, France
Supervisor: Éric Cancès

Apr 2023 – Aug 2023
Research Intern
Max Planck Institute, Numerical Methods for Plasma Physics department, Munich, Germany
Supervisor: Michael Kraus

Nov 2022 – Mar 2023

Research Intern

CEA Paris-Saclay, Thermohydraulics and Fluid Mechanics team (STMF/LMSF), Saclay, France

Supervisor: Alain Cartalade

Nov 2020 – May 2022

Research Training

CentraleSupélec, Gif-sur-Yvette, France

Supervisor: Ludovic Goudenège

Publications

2026

V-Representability Beyond Singular Times in Time-Dependent Density Functional Theory: The Hubbard Dimer Case

É. Cancès, T. Duez

Preprint — arXiv:2610.33837

2026

Geometric Time-Dependent Density Functional Theory

É. Cancès, T. Duez, J. van Gog, A. B. Lauritsen, M. Lewin, J. Toulouse

Physical Review Letters 136, 256401 — arXiv:2601.07724

2026

Geometric Theory of Constrained Schrödinger Dynamics with Application to Time-Dependent Density-Functional Theory on a Finite Lattice

É. Cancès, T. Duez, J. van Gog, A. B. Lauritsen, M. Lewin, J. Toulouse

Physical Review A 113, 062222 — arXiv:2601.07719

Talks

Nov 2026 (upcoming)

V-Representability Beyond Singular Times in Time-Dependent Density Functional Theory: The Hubbard Dimer Case

MOANSI International Workshop 2026 - Laboratoire Jacques-Louis Lions, Sorbonne Université, Paris, France

Apr 2026

Illustration of the Geometric Principle in TDDFT with a 1D Soft-Coulomb Model

REGAL Workshop on the Foundations and Extensions of Density-Functional Theory - OsloMet, Oslo, Norway

Mar 2026

Constrained Schrödinger Dynamics and the Foundations of TDDFT

96th GAMM Annual Meeting - Stuttgart, Germany

Posters

Oct 2025 — **Accurate Ground State Computations of Atoms and Ions in Extended Kohn-Sham Theory**
JuliaCon Local - CNAM, Paris

Sep 2025 — **Negative Atomic Ions In Extended Kohn-Sham Theory**
Modern Wavefunction Based Methods in Electronic Structure Theory - University of Pisa, Italy

Teaching Experience

Apr 2026 – May 2026 — **Partial Differential Equations: Variational Approaches**
École Nationale des Ponts et Chaussées — Undergraduate Year 3 — Lecture + Tutorials 12h
Reference professor: Frédéric Legoll

Jan 2026 – Apr 2026 — **Analyse des EDPs (Distributions, Fourier Analysis, L^p Spaces)**
École Nationale des Ponts et Chaussées — Undergraduate Year 3 — Lecture + Tutorials 21h
Reference professor: Virginie Ehrlacher

Sep 2025 – Oct 2025 — **Modeling with partial differential equations**
École Nationale des Ponts et Chaussées — Undergraduate Year 3 — Tutorials 7.5h
Reference professor: Tony Lelièvre

Apr 2025 – May 2025 — **Partial Differential Equations: Variational Approaches**
École Nationale des Ponts et Chaussées — Undergraduate Year 3 — Lecture + Tutorials 18h
Reference professor: Frédéric Legoll

Jan 2025 – May 2025 — **Supervision of a project of students**
École Nationale des Ponts et Chaussées — Undergraduate Year 3 — Supervision 12h

Sep 2024 – Nov 2024 — **Analyse des EDPs**
École Nationale des Ponts et Chaussées — Undergraduate Year 3 — Lecture + Tutorials 20h
Reference professor: Virginie Ehrlacher — Substitute teacher

Sep 2021 – Apr 2022 — **Convergence, Integration, Probability and PDEs**
CentraleSupélec — Undergraduate Year 3 — Tutorials 20h

Jul 2021 — **Teacher for the Summer School in Mathematics**
CentraleSupélec — Undergraduate Year 2 — Tutorials 18h

Awards

2023 — **Sophie Germain Excellence Scholarship**
Fondation Mathématique Jacques Hadamard (FMJH)
Awarded for the Master's program in Mathematics and Applications, Université Paris-Saclay / Institut Polytechnique de Paris

Schools

Nov 2026 (upcoming) — **Oberwolfach Seminar: Numerics for Evolutionary PDEs – From Nonlinear Parametrizations to Time Integration**
Mathematisches Forschungsinstitut Oberwolfach, Germany

Sep 2025 — **Modern Wavefunction Based Methods in Electronic Structure Theory**
University of Pisa - Pisa, Italy

May 2025 — **7th Mini-School on Mathematics for Theoretical Chemistry and Physics**
Laboratoire Jacques-Louis Lions - Paris, France

May 2024 — **6th Mini-School on Mathematics for Theoretical Chemistry and Physics**
Laboratoire Jacques-Louis Lions - Paris, France

Conferences Attended

Sep 2024 — **Manuel Conference - Mathematical and Numerical Analysis of Electronic Structure Models**
Stuttgart, Germany

Skills

Software — Julia, Git, C++, Python, MATLAB, LaTeX, Linux

Languages — French (native), English (C1), German (B1)

Interests

Sports — Running, Trail Running, Mountain Sports